

ActInf Livestream #026.0 ~ “Bayesian Mechanics for Stationary Processes”

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hello and welcome everyone this is the active inference lab it is active inference lab live stream number 26.0 on august 6 2021 we're going to be discussing the paper bayesian mechanics for stationary processes welcome to active inference lab everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links on this slide this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here as well today as you can find out at this short link we're going to be having the context video for the discussions on august 10th and 17th in number 26.1 and 26.2 which are going to be group discussions with some of the authors joining so if you want to join for any of these talks or discussions then please just let us know and also we have a few weeks where we don't have a paper decided so if you have a suggestion and want to be in those discussions let us know all right today in acting stream number 26.0 the goal is going to be to learn and discuss this paper bayesian mechanics for stationary processes by decosta fristen heinz and pavliotis which was posted in june 2021 this video just like all the dot zeros is just an introduction to some of the ideas in a walk through the paper one way it's not a review or a final word we're gonna give some overview to the paper then address a few of the key words from a big picture perspective then go through the figures in the formalism i'm daniel and i'm a postdoc researcher in davis california so in this paper the aims and the claims are laid out as followed they wrote in this paper we considered the consequences of a boundary mediating interactions between states internal and external to a system so it's about interfaces and interactions that's the first point on unpacking this notion we found that the states internal to a markov blanket look as if they perform variational bayesian inference optimizing beliefs about their external counterparts so uh when the boundary is set up in a certain way for which kinds of systems that's what we'll be asking uh how do those systems look from the outside or what does it look as if they're doing and then three when subdividing

the blanket into sensory and active states we found that autonomous states perform active inference in various forms of stochastic control i.e generalizations of pid control so a lot of these terms like variational bayesian markov blanket pid control we're going to be talking about more but at the overview level that's the aims and claims as the authors write them the abstract states that the paper develops a bayesian mechanics for adaptive systems and then there's only these three claims again which are again related to the blanket and adaptiveness of systems and then this partitioning of that interface into sense and action so first we model the interface between a system and its environment with a markov blanket this affords conditions under which states internals of the blanket encode information about external states second we introduce dynamics and represent adaptive systems as markov blankets at steady state this allows us to identify a wide class of systems whose internal states appear to infer external states consistent with variational inference in bayesian statistics and theoretical neuroscience finally whoops we partition the blanket into sensory and active states it follows that active states can be seen as performing active inference and well-known forms of stochastic control such as pid control which are prominent formulations of adaptive behavior in theoretical biology and engineering so this is going to be relating to math and physics with the markov blankets and the far from equilibrium thermodynamics component and information theory and then it's also going to be developed into some domain specific cases or shown to be a generalization of domain specific cases like theoretical biology and engineering these are the sections of the paper so first there's an introduction which has some of the general concepts then there's a section on markov blankets which is going to be a key framework and a key uh formalism to track because it's kind of what the paper is about so first technical sections about the markov blankets followed by a discussion of how that relates to bayesian systems or certain types of bayesian systems and then this general bayesian framework is then sort of uh specified towards an active inference model specifically with an eye towards and then there's a discussion and conclusion and then there's three supplemental sections related to the synchronization map which is about the synchronization of internal and external states and how that's related to some math results there's the helmholtz decomposition which is related to vector fields and splitting them into different components and then there's the free energy computation and some more details related to what exactly that formalism is used as here the keywords of the paper were variational bayesian inference non-equilibrium steady state markov free energy principle predictive processing and active inference so first non-equilibrium non-equilibrium steady state here's a figure from this paper stable unstable and

meta-stable states of equilibrium definitions and applications to human movement so there's going to be some mathematical definitions and uses of non-equilibrium steady state but this is a domain specific usage that gets at some of the nuances there's not just one single way to frame equilibrium especially for dynamical systems and this is shown as with an example of a person in movement so there might be a total thermodynamic like death but even within the non-death regime there are still many types of there's lying down they're sitting in a chair there's a walking steady state there's this meta-stable state and then there's unstable states of equilibrium so a few just questions would be like what does it mean to have equilibrium of a measurement or a system at a given level and what's the difference between that static just something on the ground versus a dynamic equilibrium like something that maintains its position and uses energy to do that so in those cases where does the energy come from how does this relate to dissipative so something where there's energy to be expended like a battery or a tornado and how does that relate to adaptive systems and predictive systems which ones are related to active inference one or both and what about when there are multi-level systems of equilibria so some that are higher frequency or shorter timescale or smaller spatial scale and others that are broader and also what if we want to have action-oriented models of those systems so we want to have the description that's optimal for some other set of requirements other than just accurate measurement and that's where this paper is going to be kind of headed is thinking about dynamical systems in general and looking at equations that relate to solving them another keyword was variational bayesian inference and there's a lot of good literature on this topic from a lot of different perspectives but this article had a good way of phrasing it which is variational inference methods consist in finding the best approximation of a distribution among a parameterized family and in contrary to sampling approaches a model is assumed which is the parameterized family implying a bias but also a lower variance in general variational inference methods are less accurate than sampling based like mcmc ones but produce results much faster these methods are better adapted to big scale very statistical problems so again to just kind of give a qualitative example it's like the real world is giving you this data which is the blurry image of a cat one way to get at what was the generative uh underlying cat that generated this blurry image would be to sample and find the most likely cat generating this image or do various other sampling schemes and if it's set up properly it can fit arbitrary distributions however it can be computationally expensive and really hard to know if you're solving the problem at a functional enough rate in contrast the variational inference approach is related to fitting the data based upon a pre-specified family of distributions which

can still be really general and it can be fast but it can sometimes lead to fitting the optimal support for a given distribution but having it just be categorically wrong on some other dimension like estimated through deep time or something like that so variational bayesian inference is this lower cat where there's a known cat and then you stretch and distort the cat for example reparameterize the cat to find an underlying likely cat another was markov blanket and there was good discussion of this with mel andrews in active stream 14 and we kind of had this continuum from markov on the left here which was studying systems from a mathematical perspective didn't have computer so it was looking at matrices and which elements of matrices were connected to each other or not and then that was developed later by pearl and others in this broader bayesian statistical framework that's where the markov blanket terminology starts to come from and it's also oriented towards empirical data and discovering dependencies in data and it's very computational and then part of the literature that we're discussing is where fristen and others have built on or used this pearl blanket slash markov theme and done a few things like separating the blanket states into sensory in action incoming and outgoing nodes uh introducing the cybernetic imperative to be a goal seeking or a multi-scale good regulator to maintain that non-equilibrium steady state which we'll talk about or and we did um and then also what's being explored from a formalism perspective in the paper we're discussing is how is it or under what conditions do internal states act as if they're a model of their non-local dependencies so causal dependencies in the world would be reflected by how the brain or some other computer system works so that entails this generative model of the action in the niche and also brings in the element of including the uh blanket in internal external states as part of this broader inactive embodied and cultured slash pragmatic term also in active stream 14 we had this sort of progression emerging from the structuring of the paper where there's a uh sort of vector field meets diffusion thermodynamics uh branch on the bottom with the uh focker plank and helmholtz and that's related to non-equilibrium steady state densities and then the free energy principle uh subsuming all these sort of pieces like the markov blanket being maintained through time necessitates the degenerative model of action is what the internal states have to be inferring so the useful components of the external world and then uh under free energy principle there's all these sort of related areas like active inference is that an implementation of systems under the free energy principle how is this related to predictive processing bayesian brain so we talked more about that but today we're going to look at how it's used more by the authors in this paper so the first thing is that the code is available on connors github and we'll also be talking with connor hopefully so let's um

maybe work through it or if anyone has the chance to run it tell us about how it went or what they learned from it but i didn't get to run it so we can start with figure one which is laying out the big graphical intuition of the whole paper and everything is going to be specifying uh constraints or some details or implications of a way of interpreting this intervention of the blanket states between internal and external states so we have μ the internal states b the blanket states and η which are looks like a curly n and um this is going to allow us to separate any set of nodes μ that we're specifying and ask what are the blanket states that insulate from a some statistical codependency sense which we'll be exploring insulates these two partitions and the caption says the mark called blanket is depicted graphically as an undirected graphical model also known as a markov random field the circles represent random variables the lines represent conditional dependencies between random variables the markov blanket condition means that there is no line between μ internal states and η the external this means that μ and η are conditionally independent given b so conditioned on the blanket these internal external states are separated in other words knowing the internal state μ does not afford additional information about the external state η when the blanket state b is known thus blanket states act as an informational boundary between internal and external states so that's what the blanket states do and this paper is going to explore some of the consequences of that and again anyone who has more expertise in this area or some piece of knowledge that i totally missed or got wrong my apologies but um just want to try to represent it as i saw it because a lot of the details i was just wanting to learn more about and go over with the authors uh so if anyone has comments in the chat they can also just like mention it okay so let's jump into how they present some of the key pieces of the formalism because that's the details of free energy principle slash some of the grounding of active inference so even if this is new uh formalism for you then also hopefully it's also interesting so keeping that picture in mind of figure one another way of writing out that graphical model with a formalism is this η and then upside down t so that's the external states and then μ conditioned on b so that is blanket states are markov blanket separating internal external states so the top third of the slide these are like writing out in pros the same thing as this graphical model and then we can take that tuple that set of those three kinds of nodes so we have measurements from all these different things or generative models of each of these different categories of nodes and then that set so all the measurements slash generative that x is existing in a bigger state space so the little x which consists of like the real measurements are part of the bigger x which is kind of like the capital versions of each of these and those spaces the the uh e b and i are

taking to be euclidean spaces for simplicity so i don't know if it holds for other kinds of spaces
 so that's just defining the total space the possible states of each of these nodes and then also
 the ones that you actually observe they start with this case of a gaussian distribution p
 encoding the distribution of states of the world so p is a random variable that has drawn from n
 that's like a normal distribution symbol with a mean and a variance and the mean is zero and
 the variance is this Σ π to the negative one and Σ this has an associated precision
 covariance matrix so mean zero that's the convention n means the normal distribution and so
 it's zero mean that's the variance π and the precision matrix is π so then the variance is this π
 inverse and sometimes the equations are formulated in terms of precision other times they're
 they're discussed in terms of variance but those are kind of like two sides of the same coin
 because the same it's just like describing it whether it's how uncertain you are is like looking
 from the mean out and then your precision might be kind of like is like focusing down uh
 unpacking in terms of gaussian densities so we would find that a markov blanket is equivalent
 to sparsity in the precision matrix so the mark called blanket condition is entailing that the
 covariance the π between external states and internal states that's the first one here is
 equivalent to between the μ and the η and that both of those equal zero so there's no
 covariance and they're the same as zero so they're conditionally independent that's the
 bayesian covariance matrix framing of the markov blanket sparsity blanket states act as an
 information boundary between external and internal states and this allows the breakdown of
 instead of the all by all consideration of correlations for data fitting or generative modeling
 you can reduce the model space to a much better uh formulated subset of equations which are
 related to inferring relating to like external states conditioning on the blanket states and
 internal states conditioning on the blanket states this enables us to associate to any blanket
 state its corresponding expected external and expected internal states so then this equation said
 uh drew us a set of equivalencies which were that the internal states as a function of blanket
 states so μ of b were equal to the expectation fancy e of μ conditioned on b so that's the
 expectation of external states conditioned up on of internal states Σ internal states
 conditioned on blanket i think that's corrected is right then three that that expectation is the
 same as the distribution that's being calculated involving uh the objective distribution p and
 that also that that's related to uh the variance in covariance like this Σ variable of internal
 and blanket states μ and b mapping to the space of all the possible which was i from here
 and then they wrote pursuing the example of the nervous system each sensory impression on
 the retina and ocular motor orientation blanket state is associated with an expected scene that

caused sensory input which is the expected external state and does an expected pattern and an expected pattern of neural activity in the visual cortex expected internal state so it's a way that that objective of uh p gets integrated into the internal model's output given blanket states or maybe there's another interpretation too so i was just i don't know one way to look at it this question about the relationship between the internal and the external states after you've not denoised them but decorated them in a specific way is this topic of the synchronization map so they go from just pointing out that okay we've decorated internal and external states with these blanket now we're going to ask how these internal states might encode information about expected external so after we've decorated systems how do they have a memory or a model of one another for this we would need to characterize a synchronization function σ mapping the expected internal state to the expected external state given the blanket state so σ mapping function of internal states of blanket states is going to relate to the external states at a conditioned on blanket states so there's still the dependence of both internal and external states on the blanket states b and now there's a σ mapping that is gonna connect internal and external and lemma 2.1 goes through some details of that specification so the first is proposing that there is a function that maps for any blanket state i don't know how generally it holds but or this would be cool to talk to the authors about but it's sort of proposing this σ function and maybe in certain cases it's a zero line maybe in other cases there's some meaningful relationship but for biological systems it does seem like there's a meaningful relationship so those are the systems that we want to explore modeling they then write in how to construct this synchronization map the key idea is to map an expected internal state μ of b to an expected external state η of b and to do this the two points they provide are one to find a blanket state that maps to expected internal state i.e by inverting μ and 2 from this blanket state find the corresponding expected external state i.e by applying η we now proceed to solving this problem given an internal state μ we study the blanket states b such that the internal states of b equals μ they provide some details maybe that is meant to be external states i'm not sure um here um they're framing that inference that μ is doing the internal states in terms of covariance matrices which is in this equation 2.4 and then frames that matrix of unknowns as a system of linear equations or those are kind of similar representations and then solves for it in the vector space using equation 2.5 they use this matrix negative function which is the more penrose pseudo inverse it would be helpful to have someone explain in a little more detail what the matrix inverse is or what degrees of freedom there are in choosing different kinds of matrix inverses but i just found it interesting from wikipedia that a common

use of the pseudo inverse is to compute a best fit or least square solution to a system of linear equation that locks a solution another use is to find the minimum norm solution to a system of linear equations with multiple solutions so this means that there might be multiple solutions that are equally well fit and this just allows you to pick one fast maybe and then they write that the key for the existence of a function sigma mapping expected internal states to expected external states given blanket states is that for any two blanket states associated with the same expected internal state these be associated with the same expected external state this non-degeneracy means that the internal states eg patterns of activity in the visual cortex have enough capacity to represent all possible expected external states eg 3d scenes of the environment that's formalized in this lemma so first was the point discussed before that we're interested in positing the existence of this sigma function and then 0.2 so these are all points that are going to be equivalent to each other 0.2 is that for any two blanket states one and two that the if the two external states are equal i'm sorry if the two internal states are equal the muse of those two b's then it's going to map onto equal points in external world so the blanket is a is a map that conserves certain properties of the relationship between then points three and four of the lemma 2.1 use this kernel in image notation so here was a image that was kind of about how spaces were mapped to each other so someone could definitely help explain what some of these mappings mean or what the emblem implications are and then this sideways u shape is the proper subset symbol so that's meaning that every element of the first one is in the second one but b is a bigger subset so certain mappings between the internal and blanket covariance and the external and blanket covariance and then uh similar for the pi figure 2 provides a visualization of some simulations drawn that uh support just from a simulation perspective the existence of a useful synchronization map function so figure two on the left side shows the external states given blanket states so those are the orange yeah external states up these are the flipped sorry about that so the blue are the internal states the sigma mapping corresponding to the internal states conditioned on the external and they're doing a very accurate job of mapping onto the external states in orange and then in b is a so-called non-example and i'd be curious what changed or what was different about this second example or is there more than one way to break the model what are the cases where it does or doesn't have a mapping then in section 3 they go from that sort of static estimate of variables that are just having a specific relationship to one to beijing mechanics operating through time so they write in order to study the time evolution of systems with a markov we introduce dynamics into the external blanket and internal states henceforth we assume a synchronization map under the

conditions of the lemma 2.1 so that was what was discussed up here and now in contrast with figure 1 which was just the variables by themselves now we're putting the t subscript and looking at how those variables are changing through we use bacillus to depict an intuitive example of a markov blanket that persists over time here the blanket state between the membrane and actin filaments of the cytoskeleton which mediate all interactions between the internal states and the external media external states in example 3-1 they write there are many examples of stochastic processes at a gaussian steady state p to name a few uh stationary diffusion processes with initial conditions $x_0 \sim p$ the time evolution is given by an ito stochastic differential equation appendix b next slide that will be on the next slide and then also example three one more generally any markov process at steady state p such as the zigzag process any mean zero gaussian process at steady state and any random dynamical system at steady state p so questions for people who know more or the authors would be what kinds of systems fall under this category of stochastic processes and what kind of real world systems would be modeled well or not by this and then going a little deeper into that first point in the examples appendix b points to this helmholtz decomposition and that is conditioned on this ido stochastic differential equation and again just questions for someone who knows more or for the authors what does this edo calculation entail and what does or doesn't fit under this edo framework that would work using some other definition uh just not sure about that but in this helmholtz supplement there is a very nice figure 10 that is looking at in this kind of hill climbing so in an algorithm that's looking in its local neighborhood and doing a calculation to sort of make local decision making and hill climb that action of a stochastic but hill-climbing algorithm up this gradient or down depending on which way you set it up is going to have two components to its motion this straight up the mountain component like putting a ruler on the mountain and the maximum slope and then another component that's at the same likelihood like the same altitude same energy that's going at an iso contour just like you can always do for a mountain so with that idea in mind it's possible to take the full dynamic of a stochastic trace on the top left this is like this particle that is going different colors through time that just like converging to the top of this mountain diffusing like to the bottom of the bowl like a marble going to the bottom of a bowl and then that trajectory can be separated into this spiral or i mean this um oval the time irreversible component and then a time reversible component that has a much more direct trajectory and that was in the section with remark three one which is again related to the ido stochastic differential equation which i just i'm not too sure about um but they write when the dynamics are given by this edo stochastic differential equation a markov blanket of

the steady state density does not necessarily imply that internal and external states cannot influence each other so that's i think kind of the interesting tension in question is how do certain mathematical definitions of conditional independence of blanket states separating internal external states how does that partition sometimes result in subsets or systems that don't influence each other versus systems that do so in the case of systems that don't influence each other i was thinking of like an empty bottle of air in a room of air like they're separated systems but they interact through the bottle so they could transfer heat for example through a hot bottle could transfer heat to the room but the systems would only interact conditioned on the temperature of the interface versus a person in a room could be planning or thinking about the external states so just how do we think about what kinds of systems are still having influence or a long-term implication for each other even after conditioned on the interfacing a certain way and then also just how does this example provided kind of show that that'd be good to walk through and the absence of reciprocal influences between two states in the drift sometimes but not always implies conditional independence okay going back to the main uh facts and figures in figure four there's now looking at this synchronization map and transition probabilities for processes at the gaussian steady state so on the left is we plot the synchronization map as in figure 2. only here the samples are drawn from trajectories of a diffusion process with a markov blanket so now it's a temporal process on the right panel there are the transition probabilities of that same diffusion process for the blanket state at two different times and this is called a joint distribution because there's two distributions that are being looked at jointly and if you kind of went to one side and looked through the cornfield one way you'd see this gaussian $p(b|s)$ and if you looked at it the other way you'd see $p(b|t)$ and uh then they write which is kind of um this shows that in general processes at a gaussian steady state are not gaussian processes so that was just like very i'm curious about what that meant in fact the ornstein olbeck process is the only stationary diffusion process that is a gaussian process so the transition probabilities of nonlinear diffusion processes are never multivariate gaussians not sure what that means but sounded kind of interesting all right so then uh from this sort of mapping between internal and external states through time the synchronization mapping sigma they write in the section on predictive processing we can go further and associate to each μ a probability distribution over such that each internal state encodes beliefs about external states so now rather than just framing these as two dynamical processes that track one another like oh there's this synchronization map between this random variable internally and this random variable externally they're going to

frame this q so q is going to be a probability distribution inside of μ that's over external states such that each of those states encode beliefs about external states so q of internal states is about η and that's going to be a normal with a mean of the actual external state it's like mean and variance μ so it's η and then σ of μ the mapping function comma variant so here these two are both what it's being uh is being inferred is the the external state and the mapping and then there's the variance of the precision on external states and then μ 3.4 is just giving a little bit more details about how that q is like this internal function where the states of it are reflecting beliefs about external μ and then there's a relationship to how the internal states are conditioned on blanket states this was a μ interesting remark that'd be cool to have people discuss would be uh when they wrote note a potential connection with epistemic accounts of quantum mechanics namely a world governed by classical mechanics σ conditioned at zero or i don't know if there's another way to read that symbol with three lines in 3.2 μ not sure because in 3.2 i'm not sure if there was a σ in which each agent encodes gaussian external states the could appear to the agents as reproducing many features of quantum mechanics so this was just cool like what is the relationship between the σ internal external mapping function and then classical versus quantum mechanics so maybe classical systems are like the ones where there's a clean separation between the internal and the quantum states being ones that are more uh with residual influence between so returning to the μ predictive processing they write that under that 3 4 so under this implication of the q internal expected internal states are the unique minimizer of a callback liblar divergence so that μ through time internal states through time are the minimization the argument over the μ states internal states of the d kl the kl divergence of the internal belief distribution q about q of η double line divergence between the objective of external states conditioned on so if that q distribution can converge to the p then that is being minimized effectively this measures the discrepancy between beliefs about the external world q μ sub μ of η and the posterior distribution over external variables computing the kl divergence they're going to obtain this 3.5 and μ that's argument so it's minimizing the differential uh between σ 's prediction so that's σ of μ minus the actual external states through time σ mapping function of internal states minus the actual state multiplied by so π of the precision matrix of η relating to how wrong that differential is so that's kind of an interesting framing μ and then they write that the right hand side of 3.5 that equation is commonly known as the squared precision weighted prediction error which is the discrepancy between the prediction and the expected state of the environment and it's weighted with a precision matrix that derives from the steady

state density this equation is formally similar to that found in predictive coding formulations of biological function which stipulate that organisms minimize prediction errors and in doing so optimize their beliefs to match the distribution of external so that's pretty cool and we can talk more about it but it kind of reminds me of this uncertainty reduction imperative rather than reward maximization imperative it's kind of like saying if the organisms are minimizing their not maximizing the reward they will optimize their beliefs to match the distribution of external states so maybe that's related to that variance reduction imperative um so we had the uh predictive processing section and then there's one more layer that gets added on so from the markov blanket to making it a bayesian markov blanket happening through time to introducing this q distribution that's a special type of internal state they write in the next section we can go further and associate expected internal states to the solution to the classical variational inference problem from statistical machine learning and theoretical neurobiology so it's going to be that q function is going to be a very specific kind of q so not all systems are going to be doing this subtype of variational inference or maybe they are um that is something we could discuss but we can finesse that mapping function between the internal and external states and what exactly the internal states are because if you can co-design both the internal states and the mapping then you have a lot of degrees of freedom in mapping on to external states even as they change so now we're going to do a really sub-special phrasing maybe it's a more general one but just one specific phrasing of how the internal states could map onto the external states and that's going to be as expected as the unique minimizers of a free energy functional i.e in evidence bound so now there's a f function over blanket and internal states through time and that f function is going to consist of two elements which is like a divergence term that we saw before with a divergence between q 's internal estimate of external states divergence and actual external states conditional blanket states minus the uh the evidence term and then that can be equivalent to this energy minus entropy term so that last line expresses the free energy as a difference between energy and entropy pretty cool energy or accuracy measures to what extent predicted external states are close to the true external states while entropy penalizes beliefs that are overly precise so that's cool like energy is locking in on the good solution that's like maybe the component that is going up the hill and then entropy is keeping things away from just going up the hill and keeping this other component in but if um entropy overwhelms the entropy uh the energy then the solution will not kind of fit the situation so at first sight variational inference and predictive processing are solely useful to characterize the average internal state given blanket states at steady state it is then surprising

to see that the free energy says a great deal about a system's expected trajectories as it relaxes the steady state pretty interesting so here's how they kind of show that figure five is using that free energy minimizing framework to look at variational inference and predictive processing so the highlighted part says the figure illustrates the system's behavior after experiencing a surprising blanket so kind of like a perturbation this is a multi-dimensional ornstein olbeck ou process with two external blanket and internal variables initialized at the steady state density conditioned on an improbable blanket which was given at the initial time actual distribution of x and b at time 0. and on the left top left quadrant is sort of this red to blue gradient of free energy so it's kind of high elevation to low elevation and through time the blanket stage just converged down to the bottom of the bowl so that's the free energy minimization in action the upper right is the is the free energy over time averaged over multiple trajectories so it starts out really high and then it kind of rolls down and it converges to this low level at 40 time points the lower left is showing this uh predictive processing framing of the q distribution versus the actual distribution and said at steady state from time step about 100 the predictions become accurate so that's showing how for the estimate of the parameter converges to the accurate external states under their simulation be cool to do a walkthrough and then on the bottom right is looking at the error prediction errors the covariance of so the two co errors um the evolution of precision weighted over time these are normally distributed with zero mean at steady state uh figure six is uh maybe i didn't change not sure exactly it's a similar figure it has a little bit of differences uh again showing the predictive lock in on a free energy minimizing process showing the average free energy initially having a slight increase maybe and then just decreasing through time and then also just showing that the um the relationship of the covariance um [Music] was raised up from zero and then started converging back to zero so just cool figures but cool to see what other people would think about them in figure four or i'm sorry section four we get to active inference and stochastic control so they write in order to model agents that interact with their environment we now partition blanket states into sensory and active states so for active everything that we've talked about before is like an undirected bayesian graph um it's not using that frist and blanket separation so that's where we're gonna be bringing in a lot of other uh cool formalisms and ideas so let's go from that sort of blanket one type of fiber in the blanket two now uh in order to model agents to interact with their environment we now partition blanket states into sensory and active states so the blanket state b of t blanket through c is now two subsets of sensory and action states through time so now that whole tuple the whole set that you need to infer the four kinds of states are external sense

action and internal states and now that previous example of the bacillus so here we had just one we had external and internal states mediated by one kind of node and now the external states are coupled to the sensory states that's s of t and sensory states are coupled to internal states internal states are coupled to action states and actions connect back to external states also sense and action are connected to each other but i wonder if that part is needed or i'm always wondering what topologies of connectedness different functionalities arise from and uh miguel aguilar's paper and recent uh guest stream speak to that so we take that dynamic markov blanket model from the previous sections and now split the blanket states into sense in that was this piece here now then they write intuitively in agents actions and internal states depend upon its sensations therefore we are interested in characterizing autonomous i.e active and internal so we can't control our senses directly but we can control internal states and active states so those two states we're going to think about that pair of nodes given sensory states so that is the following free energy that was brought up earlier so taking that free energy minimizing perspective on the distribution of μ internal states and blanket states as a whole before we introduce the split here then do a similar free energy equation with three variables on the sensory action and internal states which is just another way of saying the and the internal states μ and then etc etc more details that people can explain if they know better but the result that μ comes out relates to action so it'd be cool to learn more about what this equation means but then they write this is known as active inference because expected autonomous states minimize free energy crucially active inference is a well-known framework to describe and generate adaptive behavior in neuroscience machine learning and robotics see figure eight so figure eight is a similar diagram to we had before but now it's an active inference agent rather than just the free energy minimizer of the previous framing and then they write the figure illustrates a system's behavior after experiencing a surprising sensory state averaging internal variables for any blanket state this is an ornstein olbeck with two external one sensory one active and two internal variables initialized at steady state density conditioned upon an improbable sensory and also one pattern that they found was the free energy minimization here it's relatively monotonic it kind of doesn't ever bump back up it just goes down or stays the same but then it was kind of interesting they said we plot the free energy of the expected internal state averaged over multiple trajectories so this is not just one aberrant run in this example or maybe this is just one example but they wrote the average free energy does not decrease monotonically see figure five for an explanation and then um maybe we can learn more about that but that was kind of cool okay they continue on and write that

any mean stationary mean zero stationary gaussian process with exponentially decaying auto covariance function so it resembles itself less and less is an ornstein obe process which is kind of cool because in phylogenetics we sometimes would use this for let's just say the average femur length of some mammal and then you would be inferring a model that had an average value so it's kind of like a peg and then like a tether so it's like a diffusion term and then a mean so just a mean and a variance that drift through time is like the orangeteen olbeck or the oh you process sometimes known as dub's theorem so those kinds of processes which are apparently general enough to be really good model systems for studying math i guess thus if c equals a finite sum of exponentially decaying functions we can express s of t the state's sensory states as a linear function of several nested ou processes i.e an integrator chain from control theory so the integrator chain was kind of a cool idea and we'll talk about maybe applications of integrator chain in a few slides but just to continue with the math that they have so there's this u of t states through time sensory states is going to be this f function and then there's these u and these these kind of similar d equations so it says in this example f and f sub i are suitably chosen linear functions not sure what the pronunciation is but the squiggle are matrices and w are standard brownian motion thus we can see sensory states through time s t as the output of a continuous time hidden markov model whose hidden states s of t superscript i encode various orders of motion position velocity jerk etc these are known as generalized coordinates of motion in the bayesian filtering literature which we'll look at graphically in figure 9 and more generally the state process s of t and a function f need not be linear which enables to realize non-linear non-gaussian processes sst technically this follows as ornstein obe processes are the only stationary gaussian markov processes yet we emphasize that stochastic realization is not as well developed in the general case so this is u in a way saying that we're going to have like the position of the ball is the sensory data that's like the first pass u of where the ball is that's the measurement that's the sensory data then there's the variance and then there's the second derivative of that statistical moment and so if the variance is equal through time then you can just integrate away and the higher levels are zero and you've kind of locked on to the pattern after just one or two levels but there's other distributions where as you go to higher and higher distribution uh higher moments there's just more or different information so there's this concept of the integrator chain where these relationships of equations are all integrals or derivatives of each other which are kind of like going up or down in this rows of equations and then this helps use gaussian type mathematics like the ou stochastic process where a lot of the math is clean it allows you to apply these

systems of equations of well-behaved types oh you to model highly non-linear systems so this was related to some interesting discussions um uh i i will look forward to hearing what you have to say about these so-called generalized coordinates of motion and that relationship between control and then looking at nonlinear functions as relationships of integrals and derivatives leads to this discussion of the pid control so proportional integral derivative control pid is a well-known control method in engineering more than 90 of controllers in engineered systems implement either pid or pi no derivative like just proportional integral control the goal of pid control is to control a signal $s(t)$ its integral $\int s(t) dt$ with a 0 instead of a 1 and its derivative $\frac{ds(t)}{dt}$ close to a pre-specified target value so it's like i want 95 battery and i want to be a rectangle at 95 percent and it's derivative to be such and such during use this turns out to be exactly what happens when we take stochastic control of an with two with three orders of motion so that's just a three level model in this sort of infinite rows of the moments of the the three levels are just what are described here the signal the integral and the derivative when f is linear expected autonomous states control to what extent integral proportional and derivative processes $s(t)$ at zero one and two levels of derivative are from their target value of 0. so it's like if it's a linear response function then the first derivative is going to be a constant value as in the second derivative you get rid of it furthermore from f and k alone one can derive integral proportional derivative gains which penalize deviations of these three moments respectively from a target value of zero so these are highly optimizable problems and computable crucially these control gains are simple byproducts of the steady state density and the stochastic realization problems this is a cool question why restrict ourselves to pid control when stochastic control of integrator chains is available it turns out that when sensory states s of t are expressed as a function of an integrator chain one to get away with controlling an approximation of the true sensory process obtained by truncating high orders of motion as these have less effect on the dynamics though knowing when this is warranted is a problem in approximation theory so how let's just say we open the door to doing infinite derivatives of statistical infinite variances upon variances how would we know when to stop that's the problem in approximation theory or approximation science this may explain why integral feedback which is just like battery good pi control n equals one and then the pid control with two extra layers of the derivatives are the most ubiquitous control methods in engineering applications however when simulating biological biological control usually with highly non-linear systems it is not uncommon to consider generalized motion to fourth or sixth order so that's pretty cool um like how many levels of recursion and um depth in modeling are needed

to have a people so calling back to here when we defined that initial sense as part of this integrator chain setting off the integrator chain here's that initial sense function as part of a broader set of dependencies so here is that sense function at time 3 6 and 9. the figure is depicting in a graphical format the bayesian network of those equations flipped the encircled variables are random variables and the processes are indexed an arbitrary sub an arbitrary arbitrary sequence of subsequent times the arrows represent relationships of causality so the sensory states are not causing each other through time they're getting measured through time but they are being emitted through time by this s at the zeroth level like the just the base generative model of this s of zero level function so this is the generative then that can have higher derivatives of change and each of those are being framed as well behaved statistically so in the hidden markov model and these um white nodes are hidden states but they're inferred by statistics the hidden integrator state hidden state processes sat is given by an integrator chain i.e nested stochastic differential equations these processes can be seen as encoding the position velocity jerich etc so higher derivatives of the distribution so this integrator chain is kind of a cool idea so here was one paper stabilization of integrator chains in the presence of magnitude and rate saturations a gain scheduling approach so this paper wrote and the conclusions in this paper we have given a time varying controller so kind of like a heuristic or an algorithm for control for stabilizing a chain of integrators saturations it is proved that the controller gives a convergent closed-loop system i.e for any bounded state initial or initial condition the state will converge to the origin the main strength of the controller is that it gives provable convergence without being overly conservative this is verified in a simulation study it's kind of like we have the frame of the airplane at steady state and then which is flying through the air so how are we going to re-equilibrate all the sensors so that if we get knocked off to the side what's the path by which we're going to get those sensors back to their desired level and it's just a question i oh and this is kind of what it looked like one of the figures from the paper so these are these derivative functions they get get like bumped and then they can come back to the level that they're desired almost in this signal processing like way so what are integrator chains how are they used maybe someone has used them before that'd be cool and then one kind of random thought was there's a lot of cases where we frame a process as mean zero because it allows thinking about something in to uh a defined axis whether it's the number line or whether it's some other it kind of splits half the things negative and positive or there's various other times but there's actually a bunch of places and maybe more or maybe these are not all the same so it's okay if somebody knows more and helps us all out but there's

actually a few cases where a distribution that's hard to estimate is replaced with a zero mean like sometimes there's replacing elements of a list with a z score like their standard deviation from the mean or the mode μ but then it's like half or above and half are below so that's like replacing something with a zero center z score or t distribution sometimes there are error distributions which are extracted from a signal it's like there is the underlying line and the goal of the signal extraction is to leave your error mean zero and then equally distributed in both directions so sometimes we're pulling out an error distribution we're trying to find uh that mean zero get to a level of derivative of a signal where the mean is zero like where the rate of change of some high derivative is zero which implies that it's not changing at that level and then sometimes we're trying to get μ like in predictive processing study the differentiable or predictive processing with trying to get a zero centered variance on the sensory input but more about how we can study the differential between some controllable distribution and then some other distribution that we can't control and then how to make how does zero center that to determine which solutions are better or worse than each other and there's a lot of different methods there and then there was this other paper time optimal regulation of a chain of integrators with saturated input and internal planet variables in application to trajectory planning so again just kind of a open question what is the relationship between integrator chains and then this idea that we talk about a lot with action planning as inference then the conclusion this paper outlines some of the key relationships between stationary processes inference and control these relationships rest upon partitioning the world into those things that are internal or external to a statistical boundary known as a markov blanket when equipped with dynamics the expected internal states appear to engage in variational inference while the expected active states appear to be performing active inference in various forms of stochastic control the rationale behind these findings is rather simple if a markov blanket derives from a steady state density the states the system will look as if they are responding adaptively to external perturbations in order to recover the steady state conversely well-known methods used to build adaptive systems implement the same kind of dynamics implicitly so the system maintains a steady state with its environment so maybe there's i mean this isn't a totally accurate uh classification but there's some algorithms that are being connected on one side to each other so we have learning and optimization learning as inference all of machine learning etc learning and optim optimization and then variational inference being used as an optimization technique with the gradient descent by in machine learning and maybe in evolutionary systems then there's this question about estimating latent causes in the world and

connecting that to higher derivatives in the integrator chains or as just variables under consideration that's that mapping of external states which aren't directly seen and then maybe this one goes on this side but uh there's this whole other set of algorithms related to adaptive systems and these are framed variously but we have adaptive systems cybernetic systems anticipatory systems holism homeostatic slash active learning allostatic agents all these sort of biologically inspired yes drawing on far from equilibrium but here was like the flame dissipating like it's about something that just sort of gets to its free energy minimization just like gets there but with these active systems the question becomes how do we take this ongoing multi-scale inference side and connect it to this multi-scale action in the loop niche modifying also action planning biology section um one friend uh looked at these slides and then made a comment and they wrote i think it would be good to expand on the differences of the paradigms in section three which were lettered as they were in the paper a priori a posteriori estimate estimation afterwards estimation predictive processing and variational base i get the differences formally but still not at a super intuitive level so this might be good as a pedagogical exercise for ourselves so that'd be cool maybe everyone can think about that and come to their own explanation for what the difference is between these three and we'll talk about it in the dot one and dot two then just a few of the implications and bigger questions that are raised by the so in the section on interacting markov blankets they write the sort of inference we have described could be nuanced by partitioning the external state space into several systems that are themselves markov blankets such as markov blankets nested at several different scales from the perspective of internal states this leads to a more interesting inference problem with a more complex generative model it may be that the distinction between the sources systems we generally think of as engaging in cognitive inferential dynamics and simpler systems rest upon the level of structure in their generative models i.e the steady state densities that describe their inferential dynamics so that's pretty cool about maybe this framework or at least work in this level of analysis mapping just directly to nested markov and it brings up this whole discussion about what kinds of systems can we model with active inference so no one's saying the stock market is an auto-regressive model but people use auto-regressive models so what kinds of systems will we usefully be able to model with active inference without really just getting into a debate about whether it quote is active inference in the system just how are we going to use variational based methods with modern twists and developments related to cybernetics and multi-scale systems so that's like the instrumentalism side and then the realism side is what kind of systems are doing active so there's maybe even a

realism of philosophy branch and then even maybe on the applied side like it's a philosophical debate is the cell doing active inference doing free energy principle is it realizing certain constraints of nature or how would we know what um uh you know an ant is quote doing free energy minimization in its brain and then on the engineering side it could be argued that if a robot is implementing active inference code that it is actually doing active inference so this question about designing systems within the active inference framework so instead of saying i think all internet networks are active in front systems it's definitely a claim somebody could make interesting claim but it's quite another thing to design a computer network that is within an ontology of active inference then it would be fair to say that that network is quote doing active inference so that's like realism and maybe for some it's more of an engineering realism others it's a philosophical realism another implication and bigger question that arises in the paper is about temporally deep inference so they write this distinction may speak to a straightforward extension of the treatment on offer from simply inferring an external state to inferring the trajectories of external states so again moving from not just the point estimate current estimate of external state that was what was presented earlier to inferring the trajectories of external which is to say decomposing their higher orders of movement this may be achieved by representing the external in terms of its higher orders of motion by solving the stochastic realization problem by repeating the analysis above internal states may be seen as inferring velocity juror etc of the external consistently with temporally deep in the sense of a bayesian filter a special case of which is the extended kalman boosie filter so these are cool things to talk about we could talk about what is temporally deep active inference what are the ways that the model deals with internal and external categories and discrete variables how do we deal with the inference on external states and how does that relate to our action selection and then what happens when there's a feedback or um some other type of interaction with the niche the informational niche the social needs the stigma niche so or other questions people can write um well that was a sort of quick run through paper and i'm not sure if i got each section 100 correct so it will be good to have the author's perspective on a lot of pieces as well as a lot of other experts who i'm sure could have said a ton more and better about some of the math but we can offer up the same questions that we have for any paper like what might a good understanding enable what are the unique predictions and implications and the unique developments of this paper what are some of the next steps for free energy principle and active inference research what are the goals of this research and also to the participants and the authors what are they still curious about learning so thanks for participating it was a fun 26

and uh we hope that you'll join for 26.1 and 0.2 if you can or comment on the videos or whatever and thanks talk to you later bye